AN IMPROVED LOWER BOUND IN THE NONLINEAR ROTH THEOREM

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ABSTRACT. We construct, for every $\varepsilon > 0$, a subset of $\{1, \dots, N\}$ of size $\gg_\varepsilon N^{\gamma-\varepsilon}$ containing no nontrivial configuration $\{x,\, x+y,\, x+y^2\}$, where

$$\gamma \;=\; \frac{1}{2} + \frac{30\log 10 + 15\log 28 + 10\log 20 + 4\log 21 + \log 22}{120\log 145} \;=\; 0.77509\ldots$$

This improves the exponent $0.7685038\ldots$ of Younis. The construction instantiates Younis's iterated-difference framework at the modulus 145 with a chain of subsets of $\mathbb{Z}/145\mathbb{Z}$ of sizes 10, 28 followed by the period-four pattern $20, 21, 20, 22$, and every admissibility condition is verified by exhaustive finite computation. We show that the optimisation over all admissible chains at a fixed modulus is a dynamic program over forbidden-set states, and we solve it exactly, with big-integer certificates, at the moduli 65 and 85: the optimum at Younis's modulus 65 is $0.7732006\ldots$, attained only by the periodic chain of sizes $(7, 17, 10, 15)$, and the optimum at 85 is $0.7626068\ldots$. At 145 the state space is too large for the exact program and the chain above comes from a restricted search, so the exact optimum there, and the optimisation over the modulus, remain open. A short, dependency-free verification script is included.

1. INTRODUCTION

The *nonlinear Roth configuration*, so named after Bourgain and Chang [2], is the pattern

$$\{x,\; x+y,\; x+y^2\}, \qquad y \neq 0. \tag{1}$$

Peluse and Prendiville [9] obtained the first effective density bound for subsets of $[N] := \{1, \dots, N\}$ lacking (1), subsequently sharpened by the same authors [10] to the polylogarithmic bound $|A| = O(N/(\log N)^c)$ for an absolute constant $c > 0$.

In the other direction, polynomial-size constructions are the state of the art. The Behrend construction does not apply, and the first lower bounds come from the Furstenberg–Sárközy problem [3, 12]: any set free of square differences trivially lacks (1), since $(x+y^2)-x = y^2$, so every lower bound for square-difference-free sets applies verbatim. Ruzsa's digit construction [11] gives the exponent $0.733077\ldots$, its refinement by Beigel–Gasarch and Lewko [1, 7] gives $0.733411\ldots$, and Krachun [6] has recently crossed the barrier $3/4$ of Ruzsa's method with a new construction giving $0.7527964558\ldots$. Exploiting the additional freedom afforded by the middle term $x + y$, Younis [13, Thm. 1.1] obtained a set of size $N^{0.7685\ldots}$ lacking (1), which remains the best published lower bound for the integer problem. A Lean 4

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formalisation of Younis's two-set construction posted by Jones [4] instantiates it at the modulus 145 with sets of sizes 10 and 32 and reaches $0.77028\ldots$. (The finite-field model behaves quite differently: Lewko [8] has shown that any $A \subseteq \mathbb{F}_q$ of odd characteristic with $|A| \geq Cq^{5/6}$ contains the configuration, and constructs configuration-free sets of size $c\,q^{2/3}$ in certain non-prime fields. Neither the digit constructions used here, which rely on base-m expansions and m-adic valuations of integers, nor those field-theoretic constructions transfer between the two settings.)

Our results are the following.

1.1. **Theorem.** *For every $\varepsilon > 0$ and every positive integer N there exists $A \subseteq [N]$ containing no configuration (1) with $y \neq 0$, such that $|A| \gg_\varepsilon N^{\gamma-\varepsilon}$, where*

$$\gamma \;=\; \frac{1}{2} + \frac{30\log 10 + 15\log 28 + 10\log 20 + 4\log 21 + \log 22}{120\log 145} \;=\; 0.77509186393\ldots$$

The improvement over Younis's $0.76850382\ldots$ is 6.59×10^{-3} in the exponent, and the improvement over the two-set construction at 145 of [4] is 4.80×10^{-3}.

The construction lives in a framework of Younis (Theorem 2.2 below), in which a squarefree modulus m and an *admissible* sequence of subsets of $\mathbb{Z}/m\mathbb{Z}$ (Definition 2.1) produce an exponent $\gamma = \sum_{n\geq 0} \log_m |R_n|/2^{n+1}$. Write $\Gamma(m)$ for the supremum of these exponents over all admissible sequences at modulus m. Our second result determines Γ at two moduli.

1.2. **Theorem.** *One has*

$$\Gamma(65) = \frac{1}{2} + \frac{8\log 7 + 4\log 17 + 2\log 10 + \log 15}{30\log 65} = 0.7732006139\ldots,$$
$$\Gamma(85) = \frac{1}{2} + \frac{30\log 7 + 17\log 16 + 8\log 15 + 5\log 13}{120\log 85} = 0.7626068179\ldots$$

The supremum $\Gamma(65)$ is attained by the periodic chain of sizes $(7, 17, 10, 15)$ of Proposition 3.2, and every admissible sequence $(R_n)_{n\geq 0}$ at $m = 65$ attaining it satisfies $(|R_n|)_{n\geq 1} = (7, 17, 10, 15, 7, 17, 10, 15, \ldots)$.

So at Younis's modulus 65 no chain beats $(7, 17, 10, 15)$, and the exponent $0.7732006\ldots$ of that chain is beaten only by changing the modulus. The chain of Theorem 1.1 gives $\Gamma(145) \geq 0.7750918\ldots$ and we do not know $\Gamma(145)$.

Method. Younis's construction of [13, Thm. 1.1] uses a *pair* of subsets of $\mathbb{Z}/65\mathbb{Z}$: Ruzsa's 7-element set R_1 avoiding all nonzero squares, and a 17-element set R_2 whose differences avoid the *squares of the differences* of R_1, used in the alternating order R_1, R_2, R_1, R_2, ... Both sets were found with Konc's maximum-clique program [5]. His general framework permits arbitrary admissible sequences, but longer chains do not appear in [13], and the construction of [4] is again a pair. We observe that the optimisation over all admissible chains at a fixed modulus is a stationary dynamic program over "forbidden-set states", and that a change of variable (Lemma 4.2: choose the next *state* rather than the next *set*) turns its Bellman equation into a finite table of maximum-clique numbers. At $m = 65$ the table has 81,000 distinct entries and the program is solved exactly (Section 4). At $m = 145$ the state space has 2^{22} elements, and we run a restricted version of the program that explores states along good chains (Section 5). Every admissibility condition is a finite statement in $\mathbb{Z}/m\mathbb{Z}$, verified exhaustively (Section 3; Appendix A).

2. The chain construction

Throughout, m is a squarefree positive integer and subsets of $\mathbb{Z}/m\mathbb{Z}$ are identified with subsets of $\{0,1,\dots,m-1\}$. For $R \subseteq \mathbb{Z}/m\mathbb{Z}$ write

$$\Delta R := \{\, r - r' \bmod m \;:\; r, r' \in R,\ r \neq r' \,\}, \qquad \mathrm{sq}(R) := \{\, d^2 \bmod m \;:\; d \in \Delta R \,\},$$

and call

$$\sigma(R) := \mathrm{sq}(R) \cup (-\mathrm{sq}(R))$$

the *state* of R. Since $\Delta R' = -\Delta R'$ for every R', one has $\Delta R' \cap \mathrm{sq}(R) = \emptyset$ if and only if $\Delta R' \cap \sigma(R) = \emptyset$. Let $\mathcal{S} = \mathcal{S}_m$ denote the set of nonzero squares modulo m. As m is squarefree, $d^2 \equiv 0 \pmod m$ forces $d \equiv 0$, so $\mathrm{sq}(\mathbb{Z}/m\mathbb{Z}) = \mathcal{S}$ and $\mathrm{sq}(R) \subseteq \mathcal{S}$ for every R.

2.1. **Definition.** A sequence $(R_n)_{n\geq 0}$ of nonempty subsets of $\mathbb{Z}/m\mathbb{Z}$ with $R_0 = \mathbb{Z}/m\mathbb{Z}$ is *admissible* if

$$\Delta R_{n+1} \cap \mathrm{sq}(R_n) = \emptyset \qquad \text{for every } n \geq 0, \tag{2}$$

i.e. no difference of two distinct elements of R_{n+1} is congruent to the square of a difference of two distinct elements of R_n. For $n = 0$ the condition says that ΔR_1 contains no nonzero square. In particular $\Delta R_1 \cap \mathrm{sq}(R_1) = \emptyset$, since $\mathrm{sq}(R_1) \subseteq \mathcal{S}$.

In Younis's formulation [13, Thm. 1.5] the condition reads $(R_{n+1} - R_{n+1}) \cap (R_n - R_n)^2 \subseteq \{0\}$, which is (2) once differences of *distinct* elements are used. The following is his Theorem 1.5 in the case $k = 2$. We include the short proof to keep this note self-contained, and because the dynamic program of Section 4 leans on the exact form of the digit assignment.

2.2. **Theorem** (Younis)**.** *Let m be squarefree and let $(R_n)_{n\geq 0}$ be admissible. Then for every $\varepsilon > 0$ and every N there exists $A \subseteq [N]$ with no nontrivial configuration (1) and*

$$|A| \gg_{m,\varepsilon} N^{\gamma-\varepsilon}, \qquad \gamma = \sum_{n\geq 0} \frac{\log_m |R_n|}{2^{\,n+1}}.$$

Proof. For an integer $i \geq 1$ write $2^n \parallel i$ when $2^n \mid i$ and $2^{n+1} \nmid i$. For an integer $Y \geq 1$ let A_Y be the set of integers with m-ary expansion $\sum_{0\leq i<Y} u_i m^i$ whose digits satisfy

$$u_0 \in R_1, \qquad u_i \in R_n \text{ whenever } i \geq 1 \text{ and } 2^n \parallel i.$$

(Odd positions carry arbitrary digits, since $R_0 = \mathbb{Z}/m\mathbb{Z}$.)

Suppose $u, v, w \in A_Y$ satisfy $w - u = (v-u)^2$ with $v \neq u$. Then also $w \neq u$. Let j (resp. ℓ) be the least index at which the digits of w, u (resp. v, u) differ, so that

$$w - u = m^j(w_j - u_j) + z\, m^{j+1}, \qquad v - u = m^\ell(v_\ell - u_\ell) + \tilde{z}\, m^{\ell+1}$$

for some integers $z, \tilde{z}$, with $0 < |w_j - u_j|,\ |v_\ell - u_\ell| \leq m-1$.

We claim $j = 2\ell$. Write v_p for the p-adic valuation. For every prime $p \mid m$ we have $v_p(w-u) \geq j$, with equality if and only if $p \nmid (w_j - u_j)$. Since $0 < |w_j - u_j| < m$ and m is squarefree, $w_j - u_j$ cannot be divisible by every prime factor of m, so $v_{p_0}(w-u) = j$ for some prime $p_0 \mid m$. Likewise $v_p(v-u) \geq \ell$ for every $p \mid m$, with equality at some prime $p_1 \mid m$. Since $w-u = (v-u)^2$, we have $v_p(w-u) = 2\,v_p(v-u)$ for every prime p. Taking $p = p_0$ gives $j = 2v_{p_0}(v-u) \geq 2\ell$; taking $p = p_1$ gives $v_{p_1}(w-u) = 2\ell \geq j$. Hence $j = 2\ell$.

Dividing $w-u=(v-u)^2$ by $m^{2\ell}$ and reducing modulo m yields

$$w_j-u_j\equiv(v_\ell-u_\ell)^2 \pmod m,$$

and neither side vanishes modulo m: indeed $0<|w_j-u_j|<m$, while $m\mid(v_\ell-u_\ell)^2$ would force $m\mid(v_\ell-u_\ell)$ because m is squarefree, impossible for $0<|v_\ell-u_\ell|<m$.

If $\ell\geq 1$ and $2^n\parallel\ell$ then $2^{n+1}\parallel j$, so $u_j,w_j\in R_{n+1}$ and $u_\ell,v_\ell\in R_n$, and the displayed congruence exhibits a difference of distinct elements of R_{n+1} congruent to the square of a difference of distinct elements of R_n, contradicting (2). If $\ell=0$ then $j=0$ and the three digits u_0,v_0,w_0 lie in R_1, contradicting $\Delta R_1\cap\mathrm{sq}(R_1)=\emptyset$ (Definition 2.1). Hence A_Y has no configuration (1). Note that the argument never uses $w\neq v$, so the degenerate case $y=1$ is excluded as well.

Counting digit positions, $|A_Y|=|R_1|\prod_{n\geq 0}|R_n|^{c_n(Y)}$ with

$$c_n(Y)=\#\{1\leq i<Y:2^n\parallel i\}=\lceil Y/2^n\rceil-\lceil Y/2^{n+1}\rceil\ \geq\ Y/2^{n+1}-1.$$

Given $\varepsilon>0$ put $n_0=\lceil\log_2(1/\varepsilon)\rceil$, so that $\sum_{n>n_0}\log_m|R_n|/2^{n+1}\leq 2^{-n_0-1}\leq\varepsilon$. Keeping only the positions with $n\leq n_0$,

$$|A_Y|\ \geq\ \prod_{n\leq n_0}|R_n|^{Y/2^{n+1}-1}\ \geq\ m^{-(n_0+1)}\,m^{Y(\gamma-\varepsilon)}.$$

Finally take $Y=\lfloor\log_m N\rfloor$, so that $A_Y\subseteq[0,m^Y)\subseteq[0,N]$ and $m^Y>N/m$, and let $A=A_Y\setminus\{0\}$. Then $|A|\geq m^{-(n_0+1)}(N/m)^{\gamma-\varepsilon}-1\gg_{m,\varepsilon}N^{\gamma-\varepsilon}$. □

The hypothesis $R_0=\mathbb{Z}/m\mathbb{Z}$ costs nothing: no constraint in the proof ever involves an odd position as the position j, because $j=2\ell$ is even, so the same argument and the same formula hold for any nonempty R_0, and the full residue set is simply the best choice.

2.3. **Corollary.** *Let m be squarefree, $q\geq 0$, $p\geq 1$, and let $A_1,\dots,A_q,B_1,\dots,B_p\subseteq\mathbb{Z}/m\mathbb{Z}$ be nonempty. Write $(C_1,\dots,C_{q+p})=(A_1,\dots,A_q,B_1,\dots,B_p)$ and suppose that*

1. *ΔC_1 contains no nonzero square modulo m, and*
2. *$\Delta C_{i+1}\cap\mathrm{sq}(C_i)=\emptyset$ for $1\leq i<q+p$, and $\Delta B_1\cap\mathrm{sq}(B_p)=\emptyset$.*

Then the eventually periodic sequence $(\mathbb{Z}/m\mathbb{Z},A_1,\dots,A_q,B_1,\dots,B_p,B_1,\dots)$ is admissible, and Theorem 2.2 yields

$$\gamma=\frac{1}{2}+\sum_{i=1}^{q}\frac{\log_m|A_i|}{2^{\,i+1}}+\frac{1}{1-2^{-p}}\sum_{j=1}^{p}\frac{\log_m|B_j|}{2^{\,q+j+1}}.$$

When $q=0$ the wrap-around condition $\Delta B_1\cap\mathrm{sq}(B_p)=\emptyset$ follows from (1), because $\mathrm{sq}(B_p)\subseteq\mathcal{S}$.

Proof. Condition (1) is (2) for $n=0$, and (2) is (2) for every $n\geq 1$, the wrap-around condition covering the transitions from B_p back to B_1. Summing the geometric series in Theorem 2.2 gives the stated γ. □

Younis's Theorem 1.1 is the case $q=0$, $p=2$, $m=65$, $(|B_1|,|B_2|)=(7,17)$, and $q=0$, $p=1$ recovers Ruzsa's exponent $\frac{1}{2}+\frac{1}{2}\log_m|B_1|$.

3. The certificates

3.1. **Proposition.** *Let* $m = 145$ *and*

$$\begin{aligned}
A_1 &= \{0, 48, 55, 62, 69, 76, 117, 124, 131, 138\},\\
A_2 &= \{0, 10, 12, 22, 25, 27, 35, 37, 50, 52, 60, 62, 65, 67, 75, 77,\\
&\qquad 90, 92, 100, 102, 105, 115, 117, 127, 130, 132, 140, 142\},\\
B_1 &= \{0, 15, 18, 26, 29, 44, 47, 55, 58, 73, 76, 84, 87, 102, 105, 113,\\
&\qquad 116, 131, 134, 142\},\\
B_2 &= \{0, 7, 15, 38, 46, 54, 61, 69, 76, 77, 84, 91, 92, 99, 107, 114,\\
&\qquad 122, 129, 130, 137, 144\},\\
B_3 &= \{0, 9, 13, 22, 31, 35, 44, 53, 57, 66, 75, 79, 88, 97, 101, 110,\\
&\qquad 114, 123, 132, 136\},\\
B_4 &= \{0, 22, 25, 32, 35, 47, 50, 57, 60, 72, 75, 82, 85, 92, 95, 107, 110,\\
&\qquad 117, 120, 132, 135, 142\}.
\end{aligned}$$

Then the sizes are $(|A_1|, |A_2|) = (10, 28)$ *and* $(|B_1|, \dots, |B_4|) = (20, 21, 20, 22)$*, and conditions (1)–(2) of Corollary 2.3 hold with* $q = 2$*,* $p = 4$*.*

Proof. Each condition is a finite intersection test between explicitly computable subsets of $\mathbb{Z}/145\mathbb{Z}$, and all pass. The dependency-free script in Appendix A performs every check, and recomputes γ from the six sizes through Corollary 2.3, in under a second. □

Proof of Theorem 1.1. Apply Corollary 2.3 with the sets of Proposition 3.1:

$$\begin{aligned}
\gamma &= \frac{1}{2} + \frac{\log_{145} 10}{4} + \frac{\log_{145} 28}{8} + \frac{16}{15}\Big(\frac{\log_{145} 20}{16} + \frac{\log_{145} 21}{32} + \frac{\log_{145} 20}{64} + \frac{\log_{145} 22}{128}\Big)\\
&= \frac{1}{2} + \frac{30\log 10 + 15\log 28 + 10\log 20 + 4\log 21 + \log 22}{120\log 145}.
\end{aligned}$$

□

3.2. **Proposition.** *Let* $m = 65$ *and*

$$\begin{aligned}
B_1 &= \{5, 11, 16, 22, 53, 59, 64\},\\
B_2 &= \{0, 1, 6, 11, 16, 17, 22, 27, 32, 33, 38, 43, 48, 49, 54, 59, 64\},\\
B_3 &= \{1, 6, 11, 44, 49, 54, 56, 59, 61, 64\},\\
B_4 &= \{1, 6, 12, 14, 19, 25, 27, 32, 38, 40, 45, 51, 53, 58, 64\}.
\end{aligned}$$

Then $(|B_1|, \dots, |B_4|) = (7, 17, 10, 15)$*, conditions (1)–(2) of Corollary 2.3 hold with* $q = 0$*,* $p = 4$*, and the resulting exponent is*

$$\frac{1}{2} + \frac{16}{15}\Big(\frac{\log_{65} 7}{4} + \frac{\log_{65} 17}{8} + \frac{\log_{65} 10}{16} + \frac{\log_{65} 15}{32}\Big)$$

of Theorem 1.2.

Proof. The same script checks these four sets. □

4. Optimality at the moduli 65 and 85

4.1. **The dynamic program.** Fix a squarefree m. A *state* is a set $F \subseteq \mathcal{S} \cup (-\mathcal{S})$ with $F = -F$. The states form a finite set containing $\sigma(R)$ for every $R \subseteq \mathbb{Z}/m\mathbb{Z}$. For any $B \subseteq \mathbb{Z}/m\mathbb{Z}\setminus\{0\}$ let G_B be the graph on $\mathbb{Z}/m\mathbb{Z}$ in which $x \sim y$ if and only if $x-y \notin B\cup(-B)$. The cliques of G_B are exactly the sets R with $\Delta R \cap (B\cup -B) = \emptyset$; in particular $\Delta R' \cap \mathrm{sq}(R) = \emptyset$ if and only if R' is a clique of $G_{\sigma(R)}$, and $G_{\mathcal{S}}$ is the graph of Ruzsa's problem, whose cliques are the square-difference-free sets.

For a state F let $W(F)$ be the supremum of

$$\sum_{k\ge 1} \frac{\log |R_k|}{2^{\,k-1}}$$

over all sequences $(R_k)_{k\ge1}$ of nonempty sets such that R_1 is a clique of G_F and R_{k+1} is a clique of $G_{\sigma(R_k)}$ for every $k \ge 1$. Then $0 \le W(F) \le 2\log m$, and W is non-increasing under inclusion, since $F \subseteq F'$ makes every clique of $G_{F'}$ a clique of G_F. By Definition 2.1, $(\mathbb{Z}/m\mathbb{Z}, R_1, R_2, \dots)$ is admissible if and only if $(R_k)_{k\ge1}$ is such a sequence for $F = \mathcal{S}$, and comparing with Theorem 2.2,

$$\Gamma(m) = \frac{1}{2} + \frac{W(\mathcal{S})}{4\log m}. \tag{3}$$

4.1. **Lemma** (Bellman equation)**.** *For every state F,*

$$W(F) = \max_{R} \Big(\log |R| + \tfrac{1}{2} W(\sigma(R)) \Big), \tag{4}$$

the maximum over the cliques R of G_F. Moreover W is the unique real function on states satisfying (4).

Proof. Splitting off the first set, $W(F) = \sup_{R_1}\big(\log|R_1| + \frac12 \sup(\text{value of the tail})\big)$, and the tail ranges exactly over the sequences allowed from the state $\sigma(R_1)$. As there are finitely many R_1, the outer supremum is a maximum. For uniqueness, the operator $f \mapsto Bf$, $(Bf)(F) = \max_R(\log|R| + \frac12 f(\sigma(R)))$, satisfies $\|Bf - Bg\|_\infty \le \frac12\|f-g\|_\infty$ on the finite-dimensional space of functions on states, so it has exactly one fixed point. □

The maximum in (4) runs over all cliques of G_F, which is unmanageable when F is small. The following change of variable, from the next set to the next *state*, replaces it by a maximum over the finitely many states, with coefficients that are clique numbers.

4.2. **Lemma** (Target states)**.** *For states F, F' put $D(F') := \{d \in \mathbb{Z}/m\mathbb{Z}\setminus\{0\} : d^2 \notin F'\}$ and let $T(F,F')$ be the clique number of $G_{F\cup D(F')}$. Then*

$$T(F,F') = \max\{\,|R| : R \text{ a clique of } G_F,\ \mathrm{sq}(R) \subseteq F'\,\}, \tag{5}$$

and

$$W(F) = \max_{F'} \Big(\log T(F,F') + \tfrac12 W(F') \Big), \tag{6}$$

the maximum over all states F'. Consequently W is the unique fixed point of the contraction $f \mapsto B'f$, $(B'f)(F) = \max_{F'}(\log T(F,F') + \frac12 f(F'))$, and the iterates $B'^{\,k} f$ converge to W from any starting function f.

Proof. Since $D(F') = -D(F')$, a set R satisfies $\mathrm{sq}(R) \subseteq F'$ if and only if $\Delta R \cap D(F') = \emptyset$, which gives (5). For (6), let R attain the maximum in (4) and put $F' = \sigma(R)$. Then $T(F, F') \geq |R|$ by (5), so the right-hand side of (6) is at least $W(F)$. Conversely, for any state F' let R be a clique of $G_{F \cup D(F')}$ with $|R| = T(F, F')$. Then $\mathrm{sq}(R) \subseteq F'$ and $F' = -F'$ give $\sigma(R) \subseteq F'$, hence $W(\sigma(R)) \geq W(F')$ by monotonicity, and (4) yields $W(F) \geq \log|R| + \frac{1}{2}W(\sigma(R)) \geq \log T(F, F') + \frac{1}{2}W(F')$. The last assertion follows as in Lemma 4.1. □

4.2. **The computation at $m = 65$.** Since $-1 \equiv 8^2 \pmod{65}$ we have $\mathcal{S} = -\mathcal{S}$, and $-d^2 = (8d)^2$ is a square for every d. Every state is therefore a union of some of the ten negation pairs $\{s, -s\} \subseteq \mathcal{S}$, so there are $2^{10} = 1024$ states, and (6) is a finite system whose coefficients are the 1024^2 clique numbers $T(F, F')$. The forbidden sets $F \cup D(F')$ take only 81,000 distinct values. We computed each clique number exactly by a bitset branch-and-bound with a greedy-colouring bound, as in [5]. Since G_B is a Cayley graph one may assume $0 \in R$, so every instance is a maximum-clique problem on at most 64 vertices, and the whole table takes about two seconds. The clique numbers were cross-checked against an independent implementation on random entries, and the stage-one value $T(\mathcal{S}, \mathcal{S}) = 7$ is Ruzsa's.

Value iteration for (6) in floating point converges to 10^{-13} in 46 steps and produces a stationary policy π, a choice $F \mapsto \pi(F)$ of a maximising target state for every F. Floating-point arithmetic is then discarded and π is certified exactly as follows. Let $f_\pi(F)$ be the value of following π from F,

$$f_\pi(F) = \sum_{k \geq 0} 2^{-k} \log T(F_k, F_{k+1}), \qquad F_0 = F,\ F_{k+1} = \pi(F_k).$$

The state sequence is eventually periodic (at $m = 65$ every path under π enters a cycle of length four within four steps), so $f_\pi(F)$ is a rational linear combination of logarithms of integers at most 65, computed exactly. Two facts:

(a) $f_\pi \leq W$. Indeed (6) gives $W(F) \geq \log T(F, \pi(F)) + \frac{1}{2}W(\pi(F))$ for every F, and iterating this inequality (the operator on the right is monotone and contracting with fixed point f_π) yields $W \geq f_\pi$.

(b) If $f_\pi(F) \geq \log T(F, F') + \frac{1}{2} f_\pi(F')$ for all states F, F', that is $f_\pi \geq B' f_\pi$, then $f_\pi \geq W$: the operator B' is monotone, so $f_\pi \geq B' f_\pi \geq B'^2 f_\pi \geq \cdots$, and the iterates converge to W by Lemma 4.2.

Each of the 1024^2 inequalities in (b) is, after clearing the (small) denominators, an inequality between two products of integer powers of integers at most 65, and is decided exactly with big integers. In practice all but 1797 of them hold with a floating-point margin above 10^{-6}, which is far beyond the rounding error of the screen, and the remaining 1797 were decided by the exact comparison. None is violated. Hence $W = f_\pi$ exactly, and in particular

$$W(\mathcal{S}) = f_\pi(\mathcal{S}) = \frac{16\log 7 + 8\log 17 + 4\log 10 + 2\log 15}{15}. \tag{7}$$

Proof of Theorem 1.2. By (3) and (7), $\Gamma(65) = \frac{1}{2} + W(\mathcal{S})/(4\log 65)$ is the stated value, and the chain of Proposition 3.2 attains it by Corollary 2.3.

For the second statement, the exact comparison in (b) also identifies the pairs (F, F') with equality, $W(F) = \log T(F, F') + \frac{1}{2}W(F')$. Call them the optimal actions. Let $(\mathbb{Z}/65\mathbb{Z}, R_1, R_2, \dots)$ be admissible with value $W(\mathcal{S})$, and write $F_0 = \mathcal{S}$, $F_k = \sigma(R_k)$. Since $\sum_{k \geq 1} 2^{1-k} \log|R_k| = W(F_0)$ while $\log|R_k| + \frac{1}{2}W(F_k) \leq$

$W(F_{k-1})$ for every k by (4), equality must hold at every k. As $|R_k| \le T(F_{k-1}, F_k)$ by (5) and $\log T(F_{k-1}, F_k) + \frac{1}{2}W(F_k) \le W(F_{k-1})$ by (6), this forces $|R_k| = T(F_{k-1}, F_k)$ and (F_{k-1}, F_k) to be an optimal action. Following optimal actions from $\mathcal{S}$ visits 14 states in all, every such path returns to $\mathcal{S}$ after every fourth step, and along every such path the clique numbers are 7, 17, 10, 15, 7, 17, 10, 15, ... Hence $(|R_k|)_{k\ge1}$ is this sequence.

At $m = 85$ there are 13 negation pairs and 8192 states. The forbidden sets $F \cup D(F')$ take 1,920,000 distinct values, and the table of clique numbers takes four minutes. Value iteration, the exact certificate (with 33,206 inequalities decided by big-integer comparison, none violated) and the enumeration of optimal actions (38 states) proceed exactly as above and give $W(\mathcal{S}) = \log 7 + \frac{17}{30}\log 16 + \frac{4}{15}\log 15 + \frac{1}{6}\log 13$, which is (3) for the stated value of $\Gamma(85)$. Every optimal sequence at 85 has sizes $7, 16, 15$ followed by the period-four pattern $13, 16, 13, 15$, and an explicit chain with these sizes is listed in the ancillary files. □

Thus $W(\mathcal{S})$ is larger at 85 than at 65 (4.6667 against 4.5618), but the factor $1/\log m$ in (3) wins.

5. Larger moduli

The exact program needs the full table $T(F, F')$ over all pairs of states, and the number of states is 2^P with P the number of negation pairs in $\mathcal{S}$: $P = 22$ at $m = 145$ and $P = 31$ at $m = 205$ and 221. For such m we ran a *restricted* version of the program, which only ever produces lower bounds on $W(\mathcal{S})$ together with explicit chains attaining them.

The restricted program maintains a finite set Σ of states and, for each explored $F \in \Sigma$, a finite list of transitions $(R, \sigma(R))$ with R a clique of G_F. Transitions are generated by (a) enumerating up to a few thousand maximum cliques of G_F containing 0 and recording their states, (b) for each such state F' and each negation pair $\varpi \subseteq F'$, a maximum clique of $G_{F \cup D(F' \setminus \varpi)}$, that is a largest R with $\mathrm{sq}(R) \subseteq F' \setminus \varpi$, (c) for each pair ϖ a largest R with $\mathrm{sq}(R) \subseteq \mathcal{S} \setminus \varpi$, and (d) for the states on the current best chain, a largest R with $\mathrm{sq}(R) \subseteq F'$ for each of the most valuable states $F' \in \Sigma$. Value iteration over Σ, in which an unexplored state is valued at the maximum value of its explored supersets (a lower bound on W by monotonicity), gives a lower bound on $W(\mathcal{S})$ and a policy. Following the policy realises an explicit admissible chain whose value is at least the bound, because the actual next state $\sigma(R)$ of a realised transition is contained in the planned one. The states on this chain, and the best unexplored alternatives to it, are explored next, and the whole procedure is iterated until the bound stops improving. All maximum-clique computations use the branch-and-bound of Section 4, on graphs with m vertices.

Table 1 lists the outcome at the squarefree moduli that are products of two primes $\equiv 1 \pmod 4$ up to 221, together with the exact values at 65 and 85. At 265, 305 and 377 the graphs have several hundred vertices, the maximum-clique computations dominate, and twenty-minute runs of the restricted program did not complete a periodic chain, so we report no values there. Every chain in the table was verified by the script of Appendix A written from Definition 2.1. The best chain, at $m = 145$, is the one of Proposition 3.1. The value at 205 shows that the modulus which is best for Ruzsa's problem (12 residues modulo 205, [1, 7]) is not

the best for chains, and the value at 145 shows that neither is the modulus 65: Theorem 1.2 is a statement about Younis's modulus, not about the framework.

m	$\lvert\mathcal{S}\rvert$	P	method	sizes of the best chain found	exponent
65	20	10	exact	$(7, 17, 10, 15)$ periodic	0.7732006139
85	26	13	exact	$7, 16, 15$ then $(13, 16, 13, 15)$	0.7626068179
145	44	22	restricted	$10, 28$ then $(20, 21, 20, 22)$	≥ 0.7750918639
185	56	28	restricted	$(10, 34)$ periodic	≥ 0.7596097208
205	62	31	restricted	$10, 36, 25$ then $(34, 25)$	≥ 0.7702902093
221	62	31	restricted	$11, 24, 18$ then $(24, 21, 26, 20, 24, 18)$	≥ 0.7544410087

TABLE 1. Best admissible chains found at small moduli. "Exact" means the full dynamic program with the big-integer certificate, "restricted" the lower-bound search of this section.

5.1. *Remark* (Structure of the chain at 145). The first set A_1 is a maximum square-difference-free set modulo 145, and it is the set used in [4]. The set B_1 is the full preimage of $\{0, 15, 18, 26\} \subseteq \mathbb{F}_{29}$ under reduction modulo 29: the differences of $\{0, 15, 18, 26\}$ in $\mathbb{F}_{29}$ are all quadratic non-residues, so $\{0, 15, 18, 26\}$ is a clique of the complement of the Paley graph on $\mathbb{F}_{29}$, and $|B_1| = 5 \cdot 4 = 20$. The same mechanism produces the fourth set of the optimal chain at 65 (Remark 6.1), where the lift is from $\mathbb{F}_{13}$. The pre-period matters: the chain never returns to the state $\mathcal{S}$, which would force a set of at most 10 elements, and the 28-element second set is far from a maximum clique of its graph (the maximum is 32, the value used in [4]) but leaves a state from which the four-cycle is reachable.

6. REMARKS

6.1. *Remark* (Why period four at 65). The states along the chain of Proposition 3.2 consist of 5, 9, 9 and 10 negation pairs: $\sigma(B_1)$ has 10 elements, $\sigma(B_2)$ misses only the pair $\{10, 55\}$, $\sigma(B_3)$ misses only the pair $\{26, 39\}$, and $\sigma(B_4) = \mathcal{S}$. So after four steps the chain is back at its starting state, the maximum clique of $G_{\sigma(B_4)} = G_{\mathcal{S}}$ is Ruzsa's 7, and restarting the cycle is not a choice but the only possibility at that state. The two nine-pair states differ only in which pair is free, and this determines the very different clique numbers 10 and 15. The pair $\{26, 39\}$ consists of the squares of the multiples of 13 ($13^2 \equiv 39$, $26^2 \equiv 26$), so $G_{\sigma(B_3)}$ frees exactly the differences in $13\mathbb{Z}/65\mathbb{Z}$ together with the non-square differences. Accordingly B_4 is the full preimage of $\{1, 6, 12\}$ under reduction modulo 13: the differences of $\{1, 6, 12\}$ in $\mathbb{F}_{13}$ are $\{2, 5, 6, 7, 8, 11\}$, the complete set of non-residues, so $\{1, 6, 12\}$ is a maximum clique of the complement of the Paley graph on $\mathbb{F}_{13}$, and $|B_4| = 5 \cdot 3 = 15$. Its differences are the multiples of 13 together with the residues that are non-squares modulo 13, whose squares exhaust $\mathcal{S}$, and this is why the chain closes. By the weights of Corollary 2.3 the cycle trades a smaller third stage for a large fourth stage, contributing $2 \log 10 + \log 15 = 7.313\ldots$ where the alternating pair contributes $2 \log 7 + \log 17 = 6.725\ldots$ on the same positions.

6.2. *Remark* (Two classes of Ruzsa sets at 65). The 1170 seven-element square-difference-free subsets of $\mathbb{Z}/65\mathbb{Z}$ fall into two orbits, of sizes 780 and 390, under the 780 affine maps $x \mapsto ux + t$ with u one of the 12 unit squares modulo 65 (a general

unit does not preserve square-difference-freeness). The stage-two clique numbers of the two orbits are 15 and 17. The orbit of size 390 contains B_1 and Younis's R_1, and also the first set of every optimal chain, since an optimal chain needs the stage-two value 17.

6.3. *Remark* (Scope and open questions). Theorem 1.2 settles the optimisation over chains of arbitrary length at $m = 65$ and $m = 85$. At $m = 145$ we only know $\Gamma(145) \geq 0.7750918\ldots$, and the exact value would need either the full table of 2^{44} clique numbers or an argument that prunes it, for instance an upper bound on W that is tight along good chains. The optimisation over m is open in the same sense, and the restricted search becomes expensive beyond a few hundred, where the graphs have several hundred vertices and the maximum-clique computations dominate. Two further directions suggest themselves. The analogous chains for $\{x, x + y, x + y^k\}$ with $k \geq 3$ can be treated by the same program with k-th powers of differences in place of squares and the discount $\frac{1}{2}$ replaced by $\frac{1}{k}$. And the construction of Krachun [6] for square-difference-free sets crosses the barrier $3/4$ that Ruzsa identified for digit constructions of the type used here, so whether the freedom of the middle term $x + y$ can be exploited within it is a natural question.

6.4. *Remark* (Verification). The construction of Theorem 1.1 depends only on the finite computation of Proposition 3.1, and Theorem 1.2 attains its value through Proposition 3.2. The appendix script re-derives every condition from the listed sets and computes the exponents from their sizes, using only the Python standard library. The exact values in Theorem 1.2 depend on the tables of clique numbers and the exact certificate, and the lower bounds of Table 1 on the restricted search. The programs that produce and check them (a self-contained C maximum-clique solver, the table builder, the big-integer certificate and the restricted search) accompany the paper as ancillary files.[1]

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Appendix A. Verification script

```
import math

def check(m, A, B):
    """A = pre-period sets, B = cycle sets (Corollary 2.3); returns gamma."""
    C = A + B
    for X in C:
        assert len(set(X)) == len(X) and all(0 <= x < m for x in X)
    SQ = {(x * x) % m for x in range(m)} - {0}
    diffs = lambda X: {(a - b) % m for a in X for b in X if a != b}
    sqim = lambda X: {((a - b) ** 2) % m for a in X for b in X if a != b}
    assert diffs(C[0]) & SQ == set()                    # condition (1)
    for X, Y in zip(C, C[1:]):
        assert diffs(Y) & sqim(X) == set()              # condition (2)
    assert diffs(B[0]) & sqim(B[-1]) == set()           # wrap-around
    q, p = len(A), len(B)
    g = 0.5 + sum(math.log(len(X), m) / 2 ** (i + 2) for i, X in enumerate(A))
    cyc = sum(math.log(len(X), m) / 2 ** (q + j + 2) for j, X in enumerate(B))
    g += cyc / (1 - 2.0 ** -p)
    print("m =", m, [len(X) for X in A], [len(X) for X in B], "gamma =", g)
    return g

A1 = [0, 48, 55, 62, 69, 76, 117, 124, 131, 138]
A2 = [0, 10, 12, 22, 25, 27, 35, 37, 50, 52, 60, 62, 65, 67, 75, 77,
      90, 92, 100, 102, 105, 115, 117, 127, 130, 132, 140, 142]
B1 = [0, 15, 18, 26, 29, 44, 47, 55, 58, 73, 76, 84, 87, 102, 105, 113,
      116, 131, 134, 142]
B2 = [0, 7, 15, 38, 46, 54, 61, 69, 76, 77, 84, 91, 92, 99, 107, 114,
      122, 129, 130, 137, 144]
B3 = [0, 9, 13, 22, 31, 35, 44, 53, 57, 66, 75, 79, 88, 97, 101, 110,
      114, 123, 132, 136]
B4 = [0, 22, 25, 32, 35, 47, 50, 57, 60, 72, 75, 82, 85, 92, 95, 107,
      110, 117, 120, 132, 135, 142]
g145 = check(145, [A1, A2], [B1, B2, B3, B4])           # 0.7750918639366062

C1 = [5, 11, 16, 22, 53, 59, 64]
C2 = [0, 1, 6, 11, 16, 17, 22, 27, 32, 33, 38, 43, 48, 49, 54, 59, 64]
C3 = [1, 6, 11, 44, 49, 54, 56, 59, 61, 64]
C4 = [1, 6, 12, 14, 19, 25, 27, 32, 38, 40, 45, 51, 53, 58, 64]
g65 = check(65, [], [C1, C2, C3, C4])                  # 0.7732006139958438
```

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